HKU MAIB7002 / GROUP Q

FRESH FOOD. SMARTER DISCOUNTS.



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1 Problem

Discount tomorrow?
How much is enough?

Fresh-food managers must sell perishables before they spoil, while protecting value from unnecessary markdowns.

Our goal: recommend a supported daily discount.

Data Cleaning

FreshRetailNet records daily sales, discounts and stockouts. We choose regular sellers with repeated price changes.

50,000 histories → **16,684** eligible → **312** histories / 5 stores

15-85% days at rate <0.95; 6+ state switches; <20% zero-sales days. Take the five stores with the most eligible histories.

Check duplicates, gaps and missing values; cap rates >1 and mask zero-price days. Build past-only inputs; drop incomplete history rows.

25,896 train / validation rows + 2,184 final-week cases

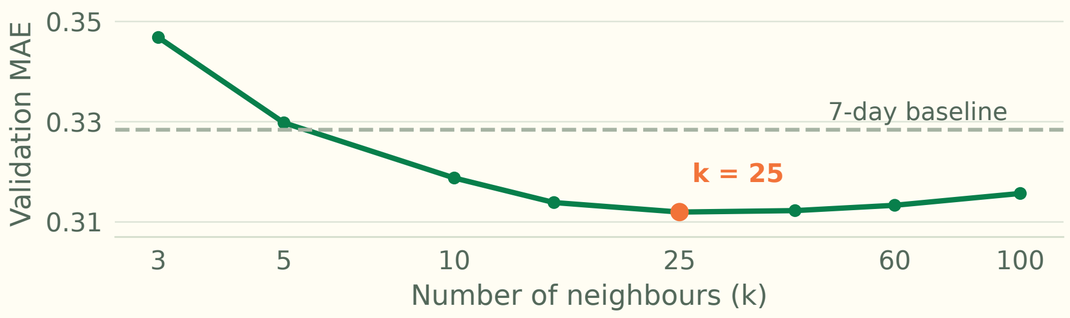
2 Models

Phase One: predict tomorrow's sales

10 decision-time inputs; five expanding validation weeks; MAE (mean absolute error) selects settings.

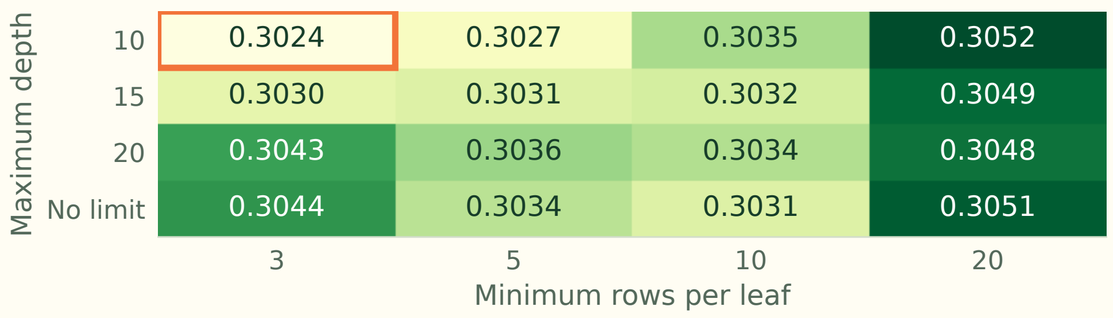
7-day mean baseline / kNN / Random Forest / CatBoost / RF + CatBoost

kNN: choose the neighbourhood



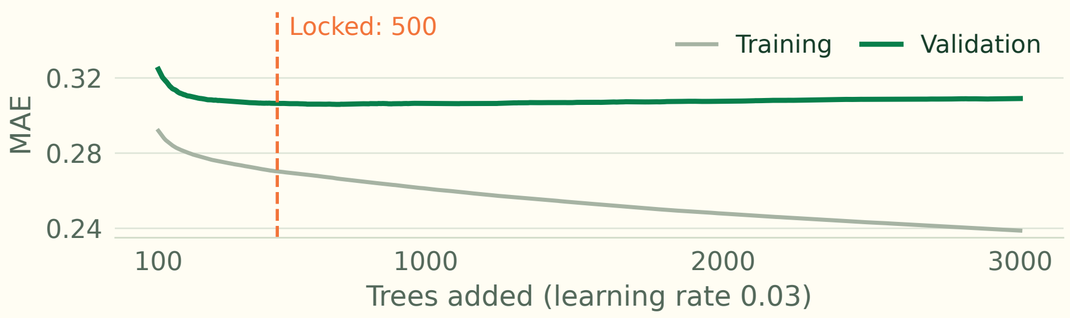
25 neighbours, standardized inputs, equal weights. The validation curve is flat near k=25.

Random Forest: limit complexity



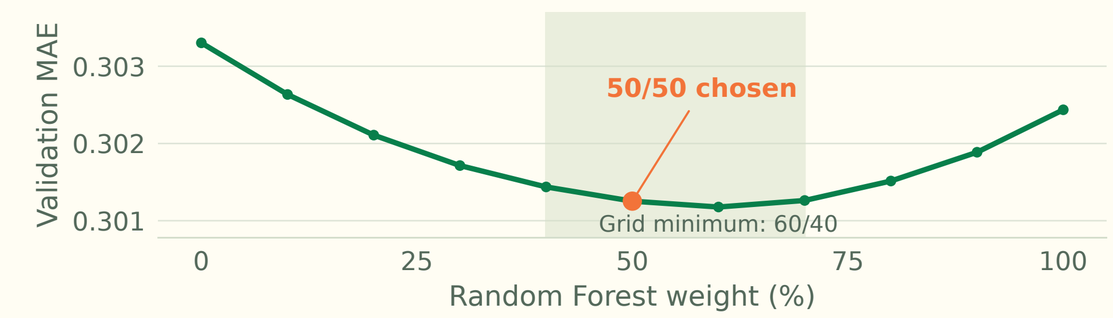
300 trees, depth 10, minimum leaf size 3. The whole tuning grid spans less than 0.003 MAE.

CatBoost: stop before overfitting



500 trees, depth 5, learning rate 0.03. On one fold, validation error turns up as training error keeps falling.

RF + CatBoost: choose simple weights



50/50 is a simple choice on a flat curve; 60/40 has the lowest tested MAE. Similar errors limit the extra gain.

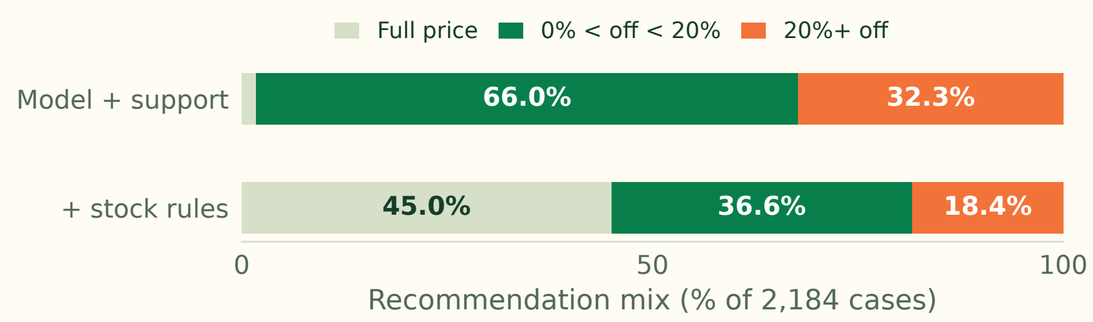
7-day mean	kNN	Random Forest	CatBoost	**50/50 ensemble**	**Final-week MAE**
0.3142	**0.3040**	**0.2923**	**0.2896**	**0.2892**	**7.9% lower** vs. the 7-day mean

3 Results & Implications

Phase Two: choose tomorrow's discount

Compare eight discounts (0-40% off). Value proxy = predicted sales x price rate; zero cost assumed.

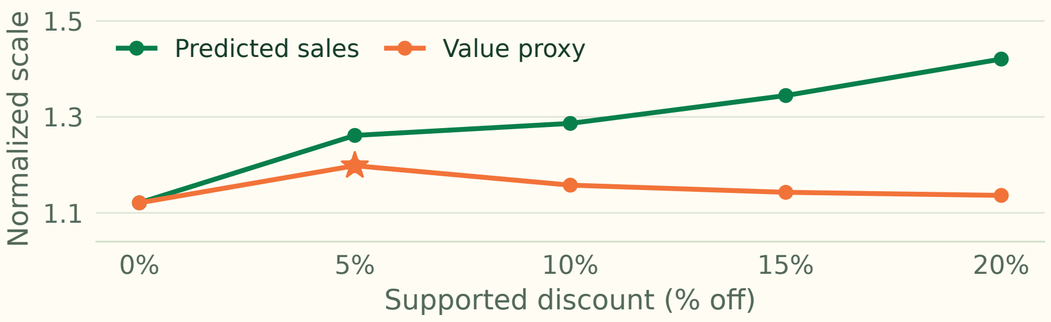
Whether: discounting becomes selective



Discount advice falls from **98.3% to 55.0%**. Stockout today (1+ hour): full price. No stockout in seven days: supported discount, if available. Otherwise, compare values.

Support: 3+ training days within 0.025; cost and gain hurdle = 0.

How much: 5% off (price rate 0.95)



Store 18 / Product 11 / 26 Jun: **5% off** predicts sales **1.261** and value **1.198**. A deeper 20% discount predicts more sales, but less value (1.136).

This product has no support for 25%, 30% or 40% off.

MANAGER TAKEAWAY

Review stock first, then choose a supported markdown that balances predicted sales and value. Set cost and gain thresholds before a controlled store pilot.

26 Jun - 2 Jul 2024: reused benchmark. Five stores in one city; retrospective selection. Stock rules are assumptions about policy.

Normalized sales and model scenarios do not prove revenue or waste gains. Source: FreshRetailNet-50K (CC BY 4.0); final notebook a7e26dd.